Introducing a Four-Practice Instructional Model for "Flipped Classrooms"*

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*Jia-Ling Lin, Paul Imbertson, Tamara Moore, Conference Proceedings, 120th ASEE Annual Conference and Exposition, 2013; 121st . ASEE Annual Conference and Exposition, 2014. References therein.

Background

In light of statistics showing shrinking enrollment in undergraduate electric energy system curricular, the power programs nationwide have reacted responsively. The Department of Electric and Computer Engineering at the University of Minnesota Twin Cities has provided a comprehensive plan to revamp teaching and learning of electric energy power systems. The STEM Education Center is collaborating with the course instructor to design and implement the instructional strategies for the new curriculum, which replace traditional lectures by problem-centered teaching and learning. It is important that we assess the effectiveness of the newly designed Four-Practice Instructional Model in "flipped classrooms" to validate pedagogies and sustain the model.

Design and Method

Applying Design-Based-Research (DBR) to intertwine the three goals of research, design, and pedagogical practices in this study.

- Develop an instructional framework and employ iterative cycles to refine theories that are used to improve instructional practices.
- Maintain close collaborations between researchers and instructors through weekly meetings and classroom observations to achieve concurrent advancement of research and instructional strategies based on evidence of the research study.

Data collection from multiple courses including:

- Recorded and transcribed students' verbal discourse in group discussions
- Two on-line surveys at the beginning and the end of the semester to assess students' efforts and readiness for non-traditional teaching and learning
- Two focus group meetings at the end of the semester to get students' perspectives and to evaluate learning grains in non-traditional "flipped classrooms"

Data analyses:

- Developing coding schemes: adapting the revised Taxonomy of Educational Objectives
- Applying event history approach: examining dynamics of classroom discourse and its impact on learning

Purpose and Research Questions

To develop an innovative instructional strategy for widespread dissemination of core courses in science-technology-engineering-mathematics (STEM) as a Model in STEM education. It is guided by the following questions:

An Overarching Research Question:

What impact, if any, does the new instructional model have on student learning?

Study (I): *In what ways does the instructional model empower instructors?*

Study II: *In what ways do pedagogies enhance learning?*

In what ways does the role of the instructor change in a non-traditional lecture hall?

Co-constructing knowledge through joint efforts of a learning community

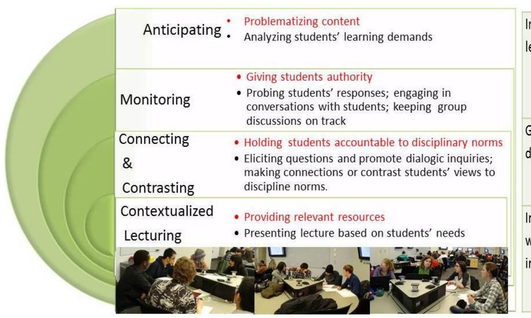
Establishing classroom discourse to support communicative teaching and learning

Coding Scheme: Knowledge Type and Discourse Structure

	Utterance structure	Cognitive process relevance
Factual	Asking questions	Remember, Understand
Conceptual	Explaining	Understand
	Exploring	Analyze, Evaluate, Create
Procedural	Critiquing/Challenging	Evaluate
	Conscious referencing	Apply, Analyze
Metacognitive	Cumulative	Remember, Apply, Create
	Key inquiry	Understand, Analyze, Evaluate, Create

Results and Main Findings

Four-Practices Instructional Model *Students' Reflections on Learning Gain*



Improved learning	*"One (thing that) I learned better, --- that I was trying to relate equations to the physical engineering problems we were talking about. I am trying to work, --- trying to understand. I looked at equations trying to think why is this way and how it is related back to the machine. I think that I was starting thinking that way."*
	"--- Drawing a picture and thinking of a physical picture, before writing down the equations and getting numbers."
Group learning dynamics	*"My role has not changed in terms as far as my motivation and my approach. At the end, I felt comfortable asking questions, trying not be worrying about asking questions, not a dumb question. Not have as much hesitations."*
	"I think there was a strong link for our group in our friendships, which lead to better attendance and eventually more success. Using that line of thinking,"
Interactions with the instructor	*"Great, I really enjoyed it. I feel like this class added a lot of the unique value and experiences. Anyone can find a ton of great lectures and learning material online for free or low cost. I enjoyed having more interactive experiences like this one."*
	"---(the instructor) was the example that how (he) used a lot ways--- he tried--- to teach the class. I learned by example. How he talked about problems, ---how he posted problems. Picked up a little bit of that way that how he posted the problem."

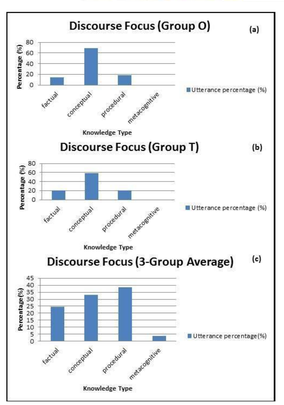


Fig.1 Utterance counts along the knowledge dimension.

- Noticeable shift of discourse focus, from factual to conceptual and procedural throughout the semester-long group work as shown in Fig. 1.
- Collaborative dialogue enables cognitive development and deep learning as shown in Figs 1 and 2.
- A learning community strengthened by the instructional model that helped to build classroom discourse.

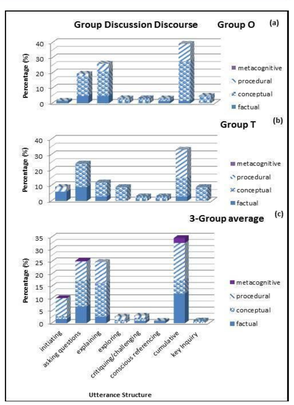


Fig.2. Dialogue counts by both the knowledge and structure

Conclusion and Future Plan

(1) The collaborative nature of dialogic discourse during group enables co-constructing knowledge and mutual responsive learning, which.
  a) changes both students' learning beliefs and behaviors considerably.
  b) helps students focus on conceptual and procedural knowledge and develop cognitive abilities for meaningful learning.

(2) The refined roles of the instructor highlight both the authoritative and responsive natures of instructors in "flipped classrooms", which
  a) is essential to the fidelity of the instructional model implementation.
  b) helps students understand what to learn and how to become an expert in the discipline.
  c) provides necessary interventions for deep learning.
  d) builds a learning community that promotes interactions and join efforts between the instructor and students as well as among students.

❑ **Next step: Conceptualizing and embedding assessment in the Instructional Model using research results.**

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